Experiment : Docker Date: 08/10/25

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Docker is a platform used to build, run and manage applications in containers.

Container is a light weight, isolated environment that contains everything your application needs to run - like:

Code, Runtime, Libraries and dependencies.

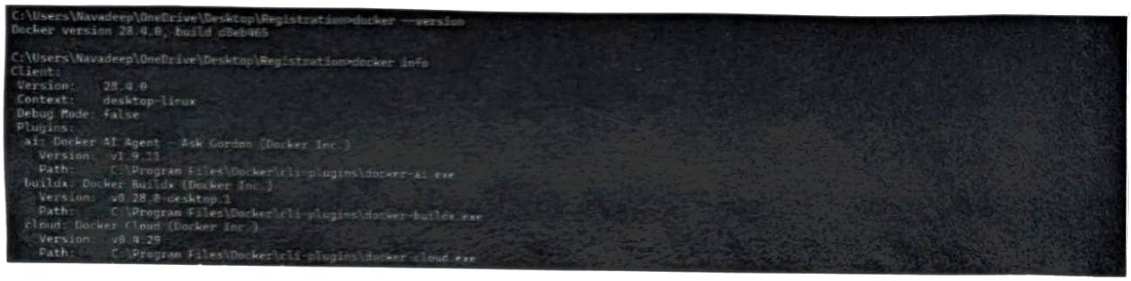
Docker Commands:

1) Docker Basics

docker --version: checks the installed docker version.

docker info: detailed information about docker.

docker help: get help for specific one.







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2> Working with Containers: Images:

docker pull <img> : download an image from docker hub

docker images : lists all downloaded images.

docker rmi <img>: Remove an image.

docker build -t name . : build an image from a docker file in the current directory.

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
4b3ffd0ccb52: Pull complete
Digest: sha256:59a458b76b4e8896831cd559576eac7d6cb53a69b38ba819fb26518536368d86
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker images
REPOSITORY              TAG       IMAGE ID       CREATED          SIZE
registration            1.0       28d5fe45148c   26 minutes ago   79.7MB
navadeep06/registration 1.0       28d5fe45148c   26 minutes ago   79.7MB
navadeep81/registration 1.0       28d5fe45148c   26 minutes ago   79.7MB
ubuntu                  latest    59a458b76b4e   9 days ago       117MB
hello-world             latest    54e66cc1dd1f   2 months ago     20.3kB

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker rmi ubuntu
Untagged: ubuntu:latest
Deleted: sha256:59a458b76b4e8896831cd559576eac7d6cb53a69b38ba819fb26518536368d86

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker build -t myapp .
[+] Building 3.6s (8/8) FINISHED                                                
```

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker run ubuntu

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker run -it ubuntu bash
root@87529dc6817f:/# navadeep81
```

# 4) Managing Containers

docker ps: List Running Containers

docker ps -a: List all Containers

docker stop name: Stop a running Container

docker restart name: Start a stopped Container (Restart)

docker rm name: Remove a Container.

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker ps
CONTAINER ID   IMAGE                COMMAND                  CREATED              STATUS              PORTS                                       NAMES
87529dc6817f   ubuntu               "bash"                   About a minute ago   Up About a minute                                       cranky_beaver
3419ebalad71   registration:1.0     "/docker-entrypoint.…"   22 minutes ago       Up 22 minutes       0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   registration-c
ontainer

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker ps -a
CONTAINER ID   IMAGE                COMMAND                  CREATED              STATUS                          PORTS                                       NAMES
87529dc6817f   ubuntu               "bash"                   About a minute ago   Up About a minute                                                   cranky
_beaver
1217b4ca832a   ubuntu               "/bin/bash"              2 minutes ago        Exited (0) 2 minutes ago                                            great_
ramanujan
3419ebalad71   registration:1.0     "/docker-entrypoint.…"   22 minutes ago       Up 22 minutes                   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   regist
ration-container
e6c5314bcbd5   hello-world          "/hello"                 41 minutes ago       Exited (0) 41 minutes ago                                           eager_
margulis
```

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker stop cranky_beaver
cranky_beaver

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker start
docker: 'docker start' requires at least 1 argument

Usage:  docker start [OPTIONS] CONTAINER [CONTAINER...]

See 'docker start --help' for more information

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker start cranky_beaver
cranky_beaver

C:\Users\Navadeep\OneDrive\Desktop\Registration>docker restart cranky_beaver
cranky_beaver
```

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker rm cranky_beaver
cranky_beaver
```

# Creating a Docker Image and Container.

1) Create a docker file and locate it to your current directory in command prompt.

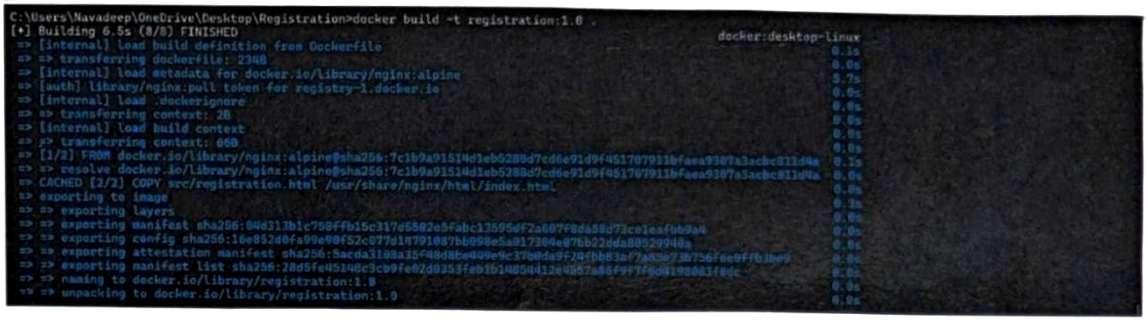
2) Build the docker image :- docker build -t registration:1.0 .

3) Verify the images :- docker images.

4) Run a Container :- docker run -d -p 8080:80 --registration

5) Verify Running Containers: docker ps

6) Test the App.













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# Uploading to Docker Hub.

1) Create a docker hub account.

2) Login to docker hub:- docker login.

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker login
Authenticating with existing credentials... [Username: navadeep81]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
```

3) Tag your local image to docker hub:-

docker tag registration:1.0 navadeep81/registration:1.0

4) Push the image to Docker Hub

docker push navadeep81/registration:1.0

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker push navadeep81/registration:1.0
The push refers to repository [docker.io/navadeep81/registration]
19a7f851f80a: Pushed
03e63548f209: Pushed
83ce83cd9960: Pushed
7fb80c2f28bc: Pushed
e2d8ea5d3690: Pushed
503ea3845ba6: Pushed
2d35ebdb57d9: Pushed
f80aba058ead: Pushed
621a51978ed7: Pushed
76c9bcaa4163: Pushed
1.0: digest: sha256:28d5fe45148c3cb9fe02d0353feb1b14854412e4557a86f9f7f6d4198083f8dc size: 856
```

5) Verify on Docker Hub.

6) Pull and Run your image anywhere

docker pull navadeep81/registration:1.0

then

docker run -p 8080:80 navadeep81/registration:1.0

```
C:\Users\Navadeep\OneDrive\Desktop\Registration>docker pull navadeep81/registration:1.0
1.0: Pulling from navadeep81/registration
Digest: sha256:28d5fe45148c3cb9fe02d0353feb1b14854412e4557a86f9f7f6d4198083f8dc
Status: Image is up to date for navadeep81/registration:1.0
docker.io/navadeep81/registration:1.0
```

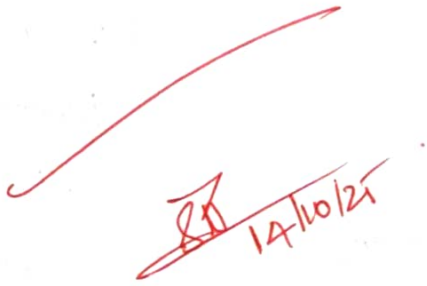
Registration From.

Name: [ ]

Email: [ ]

Phone: [ ]

[ Submit ]



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## Steps to Create a Basic Python File.

Step:01 Create a Project folder.

Step:02 Create a python script

Create a file app.py inside the folder

app.py

```
print("Hello from Python inside Docker")
```

Step:03 Create requirements.txt

requirements.txt

```
flask
numpy
```

Project folder.

↳ python-project-app

↳ app.py

↳ Dockerfile

↳ requirements.txt

Step:04 Create Dockerfile (no extension).

Docker

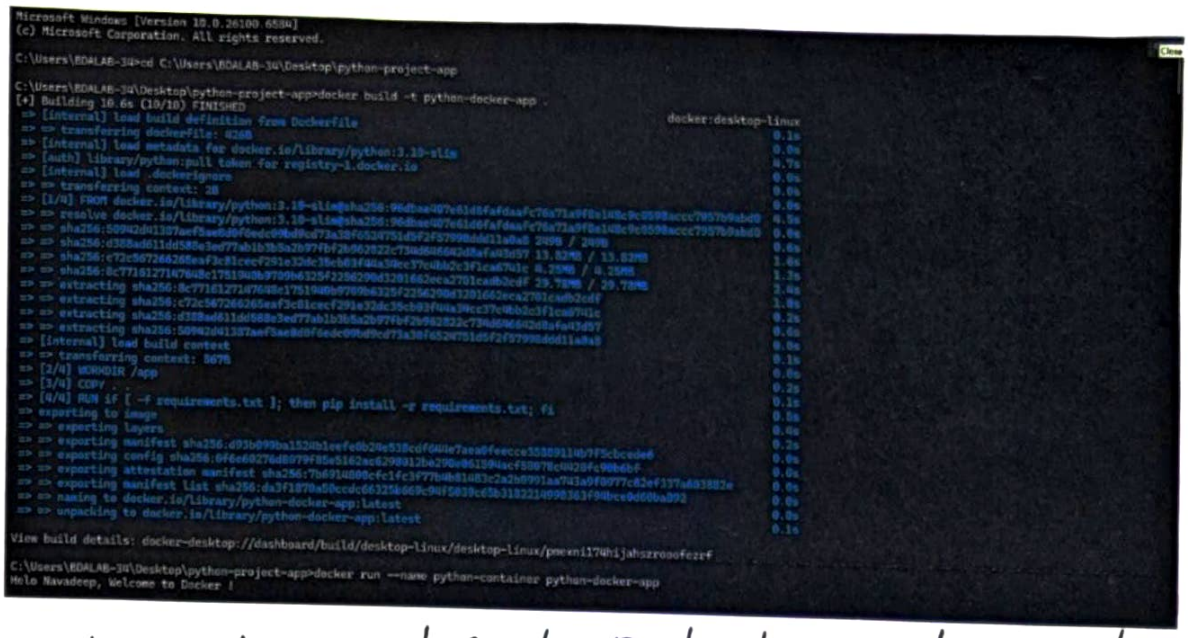
```
FROM python:3.10-slim
WORKDIR /app
COPY...
RUN if [-f requirements.txt]; then pip install -r requirements.txt; fi
CMD ["python", "app.py"]
```

Step:05 Build Docker app/Image

```
docker build -t python-docker-app
```

Step:06 Run the Docker Container

```
docker run --name python-container python-docker-app
```




01) Create a python script inside Docker to generate a random numbers of 10.

app.py

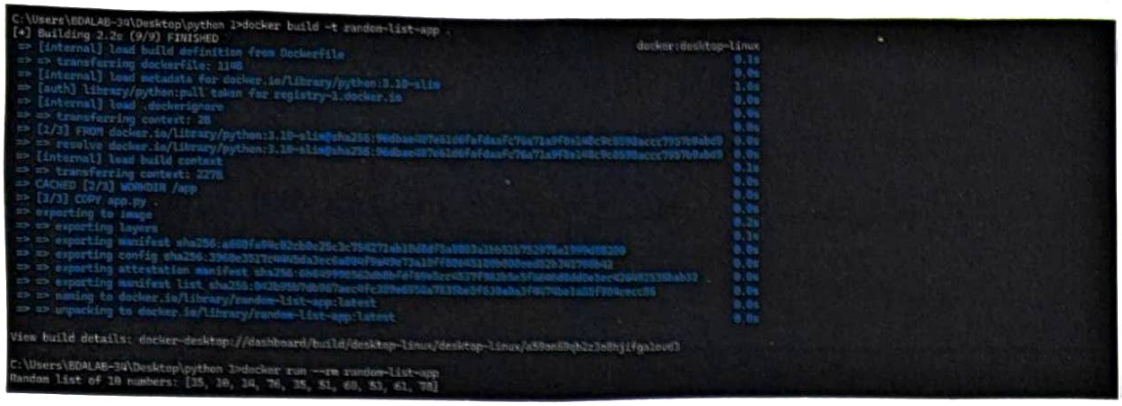
```
import random
ran_num = [random.randint(1,100) for _ in range(10)]
print("Random list of 10 numbers : ", ran_num)
```

Folder:-

random_list
↳ app.py
↳ Dockerfile.

Dockerfile

```
FROM python:3.10-slim
WORKDIR /app
COPY app.py .
CMD ["python", "app.py"]
```




Steps to Create a Basic Java File.

Step:01 Create a Project folder

Step:02 Create a java script

App.java

```
import java.util.Scanner;
import java.io.*;
class App {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a value : ");
        int a = sc.nextInt();
        System.out.println("Enter b value : ");
        int b = sc.nextInt();
        int c = a+b;
        System.out.println("The sum of two values : " +c);
        sc.close(); } }
```

Step:03 Create Dockerfile

Dockerfile

```
FROM openjdk:17-jdk-slim
WORKDIR /app
COPY App.java /app
RUN javac App.java
CMD ["java", "App"]
```

Folder:-

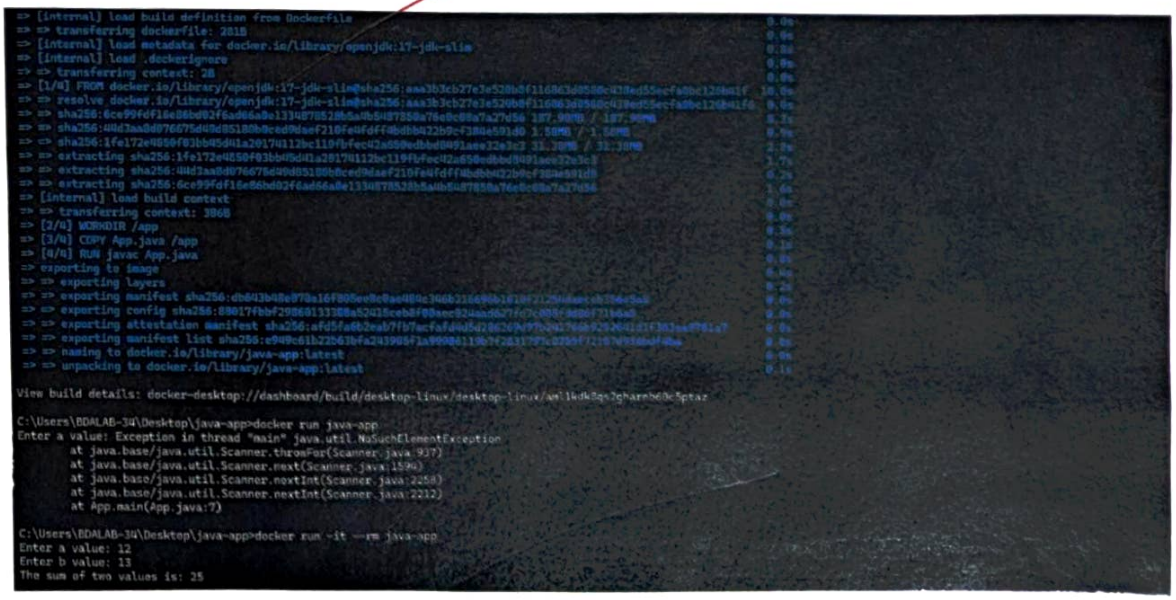
java-app
↳ App.java
↳ Dockerfile

Step:04 Build Docker image

```
docker built -t 'java-app
```

Step:05 Run Interactively

```
docker run -it java-app
```



01 Write a java program inside Docker to create Single inheritance.

App.java

```
class Animal {
  void sound() {
    System.out.println("Animal makes a sound");
    }
  }
class Dog extends Animal {
  void sound() {
    System.out.println(x: "Dog barks");
    }
  }
public class App {
  public static void main(String[] args) {
    Animal a = new Animal();
    a.sound();
    Dog d = new Dog();
    d.sound();
    }
}
```

Dockerfile

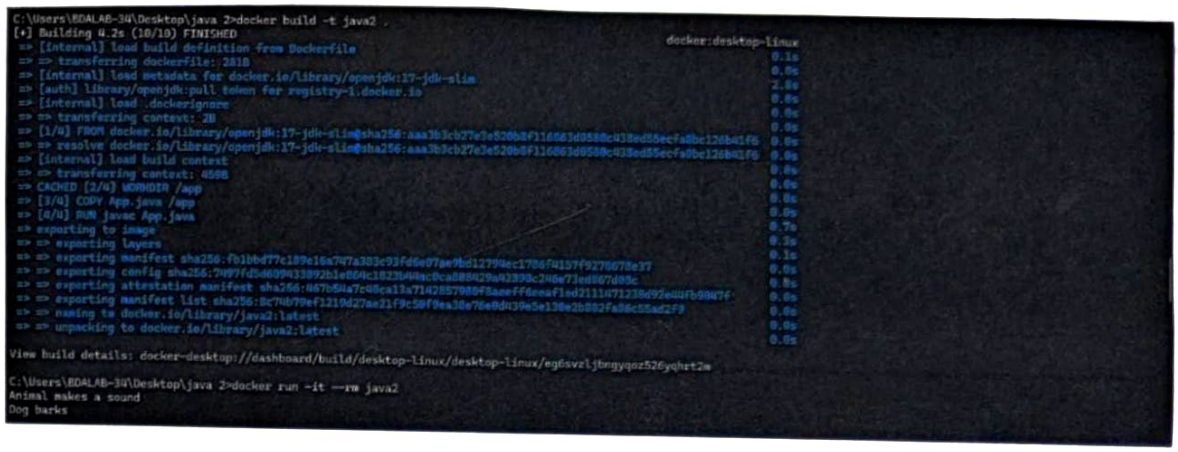
```
FROM openjdk:17-jdk-slim
WORKDIR /app
COPY App.java /app
RUN javac App.java
CMD ["java", "App"]
```

Folder:-

Single_inheritance

↳ App.java

↳ Dockerfile



```
C:\Users\BDALAB-34\Desktop\java 2>docker build -t java2 .
[+] Building 4.2s (10/10) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                0.1s
 => => transferring dockerfile: 201B                                                0.0s
 => [internal] load metadata for docker.io/library/openjdk:17-jdk-slim              2.8s
 => [auth] library/openjdk:pull token for registry-1.docker.io                      0.0s
 => [internal] load .dockerignore                                                   0.0s
 => => transferring context: 2B                                                     0.0s
 => [1/4] FROM docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438ed55ecfa0bc126b41f0  0.0s
 => => resolve docker.io/library/openjdk:17-jdk-slim@sha256:aaa3b3cb27e3e520b8f116863d0580c438ed55ecfa0bc126b41f0  0.0s
 => [internal] load build context                                                   0.0s
 => => transferring context: 458B                                                   0.0s
 => CACHED [2/4] WORKDIR /app                                                       0.0s
 => [3/4] COPY App.java /app                                                        0.0s
 => [4/4] RUN javac App.java                                                        0.9s
 => exporting to image                                                              0.7s
 => => exporting layers                                                             0.3s
 => => exporting manifest sha256:fb1bbd77c189e16a7d7a383c93fd6e07ae9bd12794ec1706f4157f9270678e37   0.1s
 => => exporting config sha256:7497fd5d6899433892b1e8564c1823b40ac0ca888429a42399c240e73ed807d89c   0.0s
 => => exporting attestation manifest sha256:467b50a7cd8ca113a7142857980f8aeeff6eeaf1ed2111d471238d92e40fb980f7   0.0s
 => => exporting manifest list sha256:8c74b79ef1219d27ae21f9c50f9ea30e76e0d439e5e130e2b882fa86c55ad2f9   0.0s
 => => naming to docker.io/library/java2:latest                                     0.0s
 => => unpacking to docker.io/library/java2:latest                                  0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/eg6svzljbngyqoz526yqhrt2w

C:\Users\BDALAB-34\Desktop\java 2>docker run -it --rm java2
Animal makes a sound
Dog barks
```

Steps to Create a Docker Compose for two python Services.

Step:01 Create a project folder

```
mkdir python-docker-compose-app
cd python-docker-compose-app
```

Step:02 Create python Scripts.

ran_num.py

```
import random
num = [random.randint(1,100) for _ in range(10)]
print("Numbers: ", num)
print("Maximum: ", max(num))
print("Minimum: ", min(num))
print("Average: ", sum(num)/len(num))
```

guess_num.py

```
import random
num = random.randint(1,50)
guess = random.randint(1,50)
print(f"Generated guess: {guess}")
if guess == number:
    print("Correct")
else:
    print("Wrong! The number was {num}")
```

Step:03 Create the Dockerfile

```
FROM python:3.10-slim
WORKDIR /app
COPY . .
CMD ["python", "random_num.py"]
CMD ["python", "-u", "ran_num.py"]
```



Step:04 Creater docker-compose.yml

```
version: '3'
services:
  ran_num:
    build: .
    container_name: ran_num_container
    command: python ran_num.py

  guess_num:
    build: .
    container_name: guess_num_container
    command: python guess_num.py
    stdin_open: true
    tty: true
```

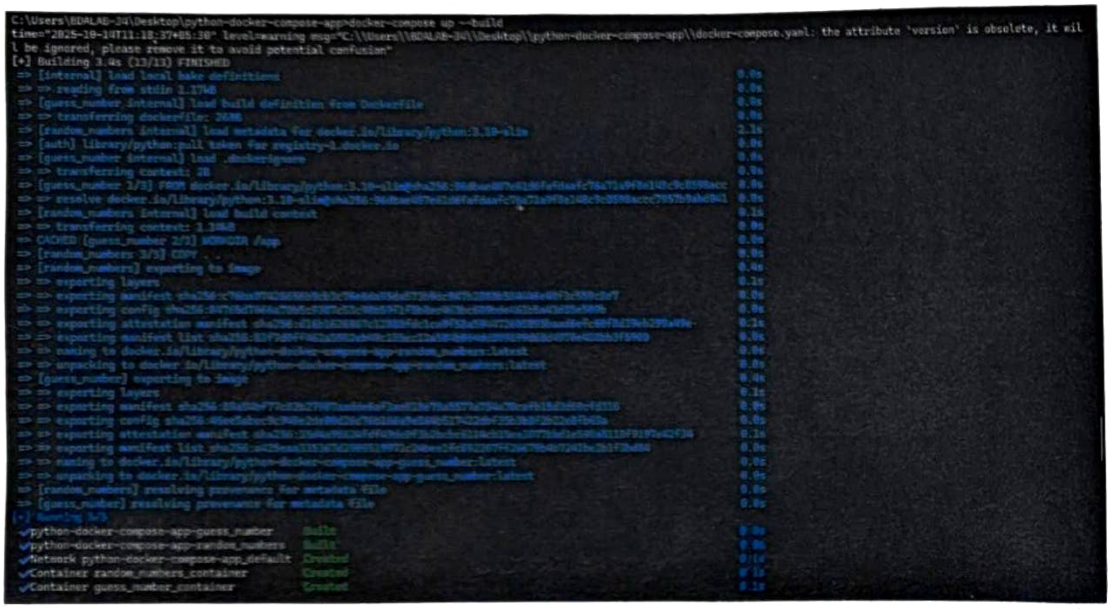
Step:05 Build and start services.

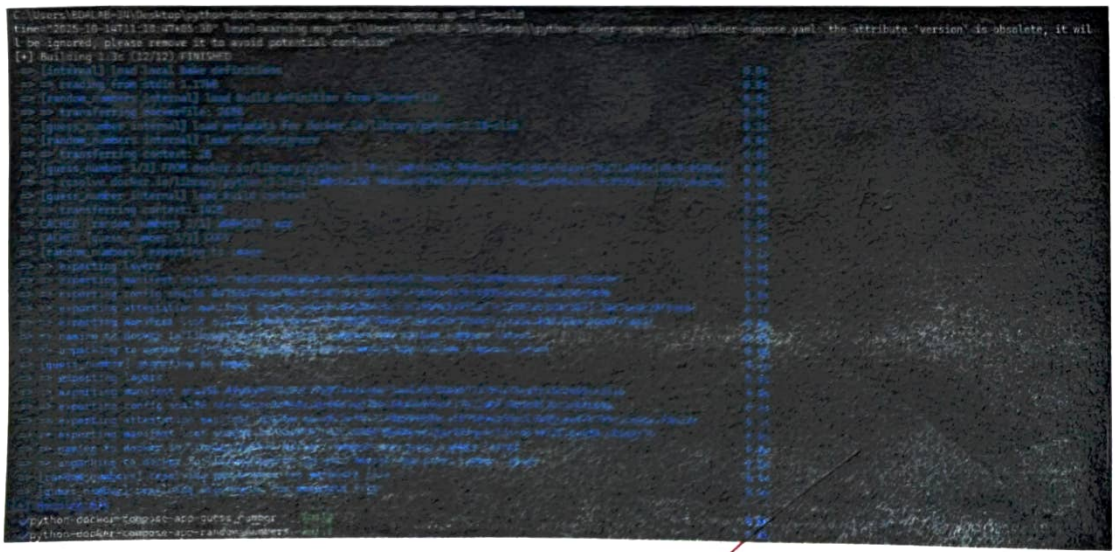
Build and start both services in foreground.

```
docker-compose up --build
```

Build and run in detached mode

```
docker-compose up -d --build.
```



Step:06 Run only the interactive service

```
C:\Users\BDALAB-34\Desktop\python-docker-compose-app>docker-compose run --rm guess_number
time="2025-10-14T11:18:54+05:30" level=warning msg="C:\\Users\\BDALAB-34\\Desktop\\python-docker-compose-app\\docker-compose.yaml: the attribute `version` is obsolete, it wil
l be ignored, please remove it to avoid potential confusion"
Generated guess: 8
Wrong! The number was 25

C:\Users\BDALAB-34\Desktop\python-docker-compose-app>docker-compose run --rm guess_number
time="2025-10-14T11:19:04+05:30" level=warning msg="C:\\Users\\BDALAB-34\\Desktop\\python-docker-compose-app\\docker-compose.yaml: the attribute `version` is obsolete, it wil
l be ignored, please remove it to avoid potential confusion"
Generated guess: 47
Wrong! The number was 45
```

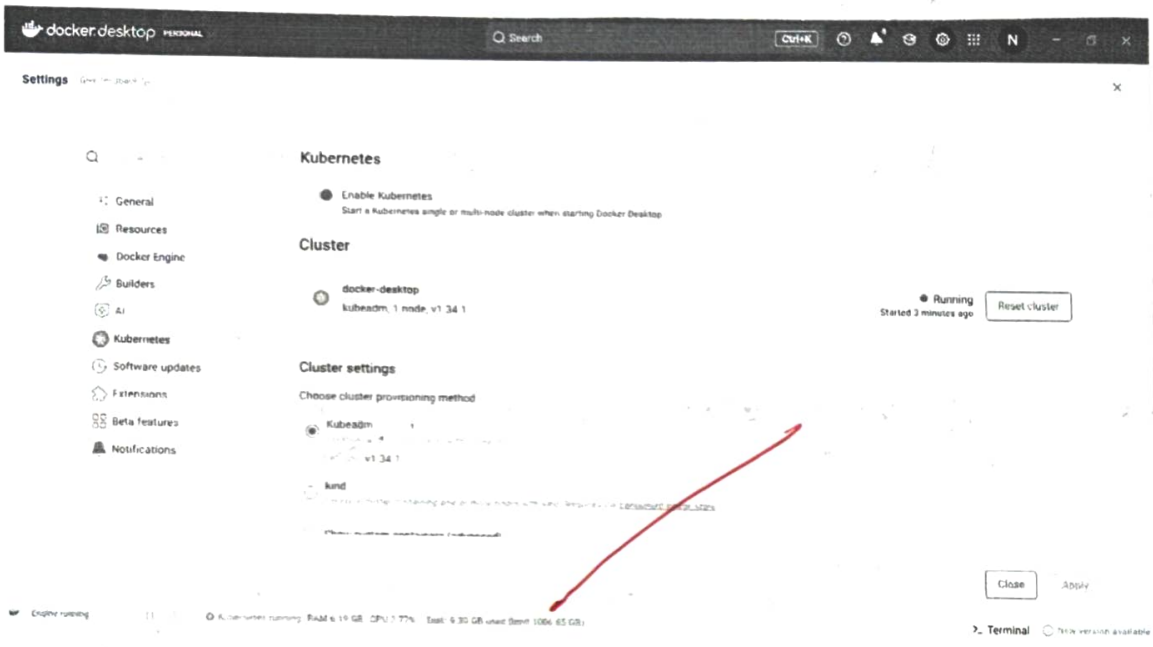
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Experiment : Introduction to Kubernetes Date:

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Kubernetes is an open-source platform used to manage and automate containerized applications. It handles deployment, scaling and monitoring automatically. It ensure applications run reliably across different environments.

Before going to perform Kubernetes, we need to enable it from Docker.




kubectl version: Check the client and server kubernetes version.

kubectl cluster-info: Displays information about the Kubernetes Control Pane

```
Microsoft Windows [Version 10.0.26200.6981]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Navadeep>kubectl version --client
Client Version: v1.34.1
Kustomize Version: v5.7.1

C:\Users\Navadeep>kubectl cluster-info
Kubernetes control plane is running at https://kubernetes.docker.internal:6443
CoreDNS is running at https://kubernetes.docker.internal:6443/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

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kubectl get nodes: List all nodes in order.

```
C:\Users\Navadeep>kubectl get nodes
NAME             STATUS   ROLES           AGE    VERSION
docker-desktop   Ready    control-plane   118s   v1.34.1
```

kubectl describe pod mypod: Displays detailed information about a specific pod.

```
C:\Users\Navadeep>kubectl describe pod mypod
Name:             mypod
Namespace:        default
Priority:         0
Service Account:  default
Node:             docker-desktop/192.168.65.3
Start Time:       Tue, 11 Nov 2025 10:34:29 +0530
Labels:           run=mypod
Annotations:      <none>
Status:           Pending
IP:
IPs:              <none>
Containers:
  mypod:
    Container ID:
    Image:          nginx
    Image ID:
    Port:           <none>
    Host Port:      <none>
    State:          Waiting
      Reason:       ContainerCreating
    Ready:          False
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-zdchv (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   False
  Initialized                 True
  Ready                       False
  ContainersReady             False
  PodScheduled                True
Volumes:
  kube-api-access-zdchv:
    Type:                    Projected (a volume that contains injected data from multiple sources)
    TokenExpirationSeconds:  3607
    ConfigMapName:           kube-root-ca.crt
```



kubectl logs mypod: View the logs from the container running in the pod.

```
C:\Users\Navadeep>kubectl logs mypod
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/11/11 05:04:49 [notice] 1#1: using the "epoll" event method
2025/11/11 05:04:49 [notice] 1#1: nginx/1.29.3
2025/11/11 05:04:49 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/11/11 05:04:49 [notice] 1#1: OS: Linux 6.6.87.1-microsoft-standard-WSL2
2025/11/11 05:04:49 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/11/11 05:04:49 [notice] 1#1: start worker processes
2025/11/11 05:04:49 [notice] 1#1: start worker process 30
2025/11/11 05:04:49 [notice] 1#1: start worker process 31
2025/11/11 05:04:49 [notice] 1#1: start worker process 32
2025/11/11 05:04:49 [notice] 1#1: start worker process 33
2025/11/11 05:04:49 [notice] 1#1: start worker process 34
2025/11/11 05:04:49 [notice] 1#1: start worker process 35
2025/11/11 05:04:49 [notice] 1#1: start worker process 36
2025/11/11 05:04:49 [notice] 1#1: start worker process 37
2025/11/11 05:04:49 [notice] 1#1: start worker process 38
2025/11/11 05:04:49 [notice] 1#1: start worker process 39
2025/11/11 05:04:49 [notice] 1#1: start worker process 40
2025/11/11 05:04:49 [notice] 1#1: start worker process 41
```

kubectl delete pod mypod: Delete the specific pod.

```
PS C:\Users\Navadeep> kubectl delete pod mypod
pod "mypod" deleted from default namespace
```

kubectl create deployment myapp --image=nginx: Create a new deployment using a container image.

```
PS C:\Users\Navadeep> kubectl create deployment myapp --image=nginx
deployment.apps/myapp created
PS C:\Users\Navadeep> kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
myapp   1/1     1            1           7s
PS C:\Users\Navadeep> kubectl get all
NAME                         READY   STATUS    RESTARTS   AGE
pod/myapp-546755b9f7-mmnkk   1/1     Running   0          11s

NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   5m50s

NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/myapp   1/1     1            1           11s

NAME                               DESIRED   CURRENT   READY   AGE
replicaset.apps/myapp-546755b9f7   1         1         1       11s
```

kubectl describe deployment myapp: Describes about the deployment

```
PS C:\Users\Navadeep> kubectl describe deployment myapp
Name:                   myapp
Namespace:              default
CreationTimestamp:      Tue, 11 Nov 2025 10:38:04 +0530
Labels:                 app=myapp
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=myapp
Replicas:               1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=myapp
  Containers:
   nginx:
    Image:         nginx
    Port:          <none>
    Host Port:     <none>
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   myapp-546755b9f7 (1/1 replicas created)
Events:
  Type    Reason             Age   From                   Message
  ----    ------             ----  ----                   -------
  Normal  ScalingReplicaSet  16s   deployment-controller  Scaled up replica set myapp-546755b9f7 from 0 to 1
PS C:\Users\Navadeep> kubectl delete deployment myapp
deployment.apps "myapp" deleted from default namespace
```

kubectl expose deployment myapp --port=80 --type=NodePort

Expose a deployment as a service to make it accessible externally.

```
PS C:\Users\Navadeep> kubectl expose deployment myapp --port=80 --type=NodePort
service/myapp exposed
PS C:\Users\Navadeep> kubectl get svc
NAME         TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)        AGE
kubernetes   ClusterIP   10.96.0.1      <none>        443/TCP        7m39s
myapp        NodePort    10.97.75.35    <none>        80:30785/TCP   5s
PS C:\Users\Navadeep> kubectl describe svc myapp
Name:                     myapp
Namespace:                default
Labels:                   app=myapp
Annotations:              <none>
Selector:                 app=myapp
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.97.75.35
IPs:                      10.97.75.35
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  30785/TCP
Endpoints:                10.1.0.8:80
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>
```

kubectl get ns: List all namespaces in the cluster.

```
PS C:\Users\Navadeep> kubectl get ns
NAME              STATUS   AGE
default           Active   8m34s
kube-node-lease   Active   8m34s
kube-public       Active   8m34s
kube-system       Active   8m34s
PS C:\Users\Navadeep> kubectl config set-context --current --namespace=default
Context "docker-desktop" modified.
```

kubectl get events --sort-by=.metadata.creationTimestamp.

```
PS C:\Users\Navadeep> kubectl get events --sort-by=.metadata.creationTimestamp
LAST SEEN   TYPE     REASON                    OBJECT                                MESSAGE
9m37s       Normal   Starting                  node/docker-desktop                   Starting kubelet.
9m36s       Normal   NodeHasSufficientMemory   node/docker-desktop                   Node docker-desktop status is now: NodeHasSufficientMemory
9m36s       Normal   NodeHasNoDiskPressure     node/docker-desktop                   Node docker-desktop status is now: NodeHasNoDiskPressure
9m36s       Normal   NodeHasSufficientPID      node/docker-desktop                   Node docker-desktop status is now: NodeHasSufficientPID
9m36s       Normal   NodeAllocatableEnforced   node/docker-desktop                   Updated Node Allocatable limit across pods
9m20s       Normal   RegisteredNode            node/docker-desktop                   Node docker-desktop event: Registered Node docker-desktop in Controller
9m15s       Normal   Starting                  node/docker-desktop
7m21s       Normal   Scheduled                 pod/mypod                             Successfully assigned default/mypod to docker-desktop
7m20s       Normal   Pulling                   pod/mypod                             Pulling image "nginx"
7m2s        Normal   Created                   pod/mypod                             Created container: mypod
7m2s        Normal   Started                   pod/mypod                             Started container mypod
7m2s        Normal   Pulled                    pod/mypod                             Successfully pulled image "nginx" in 17.6s (17.6s including waiting). Image siz
e: 59774010 bytes.
4m11s       Normal   Killing                   pod/mypod                             Stopping container mypod
3m46s       Normal   SuccessfulCreate          replicaset/myapp-546755b9f7           Created pod: myapp-546755b9f7-mnnkk
3m46s       Normal   Scheduled                 pod/myapp-546755b9f7-mnnkk            Successfully assigned default/myapp-546755b9f7-mnnkk to docker-desktop
3m46s       Normal   ScalingReplicaSet         deployment/myapp                      Scaled up replica set myapp-546755b9f7 from 0 to 1
3m45s       Normal   Pulling                   pod/myapp-546755b9f7-mnnkk            Pulling image "nginx"
3m42s       Normal   Pulled                    pod/myapp-546755b9f7-mnnkk            Successfully pulled image "nginx" in 3.291s (3.291s including waiting). Image s
ize: 59774010 bytes.
3m41s       Normal   Created                   pod/myapp-546755b9f7-mnnkk            Created container: nginx
3m41s       Normal   Started                   pod/myapp-546755b9f7-mnnkk            Started container nginx
2m44s       Normal   Killing                   pod/myapp-546755b9f7-mnnkk            Stopping container nginx
116s        Normal   ScalingReplicaSet         deployment/myapp                      Scaled up replica set myapp-546755b9f7 from 0 to 1
116s        Normal   SuccessfulCreate          replicaset/myapp-546755b9f7           Created pod: myapp-546755b9f7-mz2lt
116s        Normal   Scheduled                 pod/myapp-546755b9f7-mz2lt            Successfully assigned default/myapp-546755b9f7-mz2lt to docker-desktop
114s        Normal   Pulling                   pod/myapp-546755b9f7-mz2lt            Pulling image "nginx"
111s        Normal   Created                   pod/myapp-546755b9f7-mz2lt            Created container: nginx
111s        Normal   Pulled                    pod/myapp-546755b9f7-mz2lt            Successfully pulled image "nginx" in 3.345s (3.346s including waiting). Image s
ize: 59774010 bytes.
110s        Normal   Started                   pod/myapp-546755b9f7-mz2lt            Started container nginx
```

Experiment : _____ Date: 04/11/25

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Exp:09 Automate the process of running containerized application developed in exp:01 using Kubernetes.

Pre-requisites.

1) Docker installed and running.
2) kubectl installed.
3) Minikube (local kubernetes cluster)
4) Docker hub account.
5) Python3 and Hash library.

Step:01 Create project folder and files

mkdir simple-k8n-app

cd simple-k8n-app

Create :-

app.py

requirements.txt

Dockerfile

deployment.yml

service.yaml.

Step:02 Test app locally.

pip install -r requirements.txt

python app.py

visit https://localhost:8080




Step:03 Build docker image.




Step:04 Push image to Docker Hub.

```
C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>docker login
Authenticating with existing credentials... [Username: varshitha567]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'

Login Succeeded
```

```
C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>docker push varshitha567/simpleapp:latest
The push refers to repository [docker.io/varshitha567/simpleapp]
19cbe4e4aaf3: Pushed
e73850a50582: Pushed
38513bd72563: Mounted from varshitha567/app
3aecd199fad3: Pushed
19fb8589da02: Pushed
1e9c5fd3c8d9: Pushed
3f4677d297e4: Pushed
a9ffe18d7fdb: Pushed
6a1595b051c6: Pushed
latest: digest: sha256:0bd0fa151e44bbae993aa68a82334d4a3c6614b717d55044deef2da2bbf2ca35 size: 856
```

## Step:05 Deploy to Kubernetes

```
C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl apply -f deployment.yaml
Error from server (NotFound): error when creating "deployment.yaml": namespaces "dev" not found

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl create namespace dev
namespace/dev created

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl apply -f deployment.yaml -n dev
deployment.apps/registration-app-deployment created

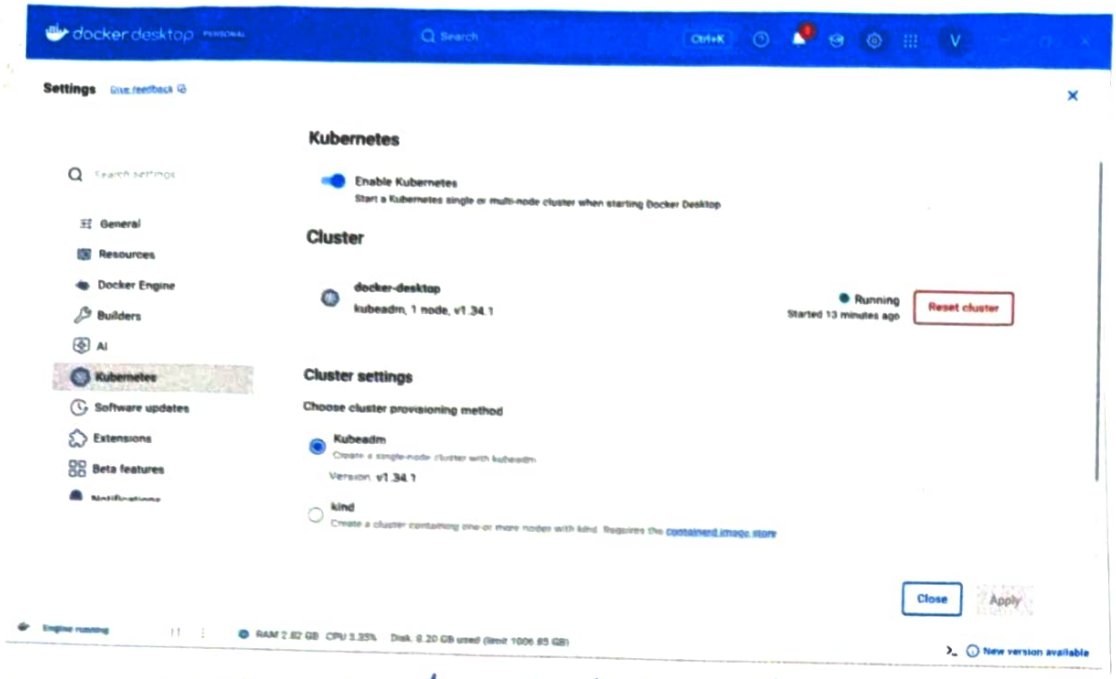
C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl apply -f service.yaml -n dev
service/registration-app-service created

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get pods -n dev
NAME                                        READY   STATUS             RESTARTS   AGE
registration-app-deployment-785b95ffc5-skt9g   0/1     ImagePullBackOff   0          19s
registration-app-deployment-785b95ffc5-smrqq   0/1     ErrImagePull       0          19s

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get svc -n dev
NAME                    TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
registration-app-service   NodePort   10.110.170.199   <none>        80:31713/TCP   18s

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get pods
NAME                                        READY   STATUS             RESTARTS   AGE
registration-app-deployment-785b95ffc5-skt9g   0/1     ImagePullBackOff   0          3m1s
registration-app-deployment-785b95ffc5-smrqq   0/1     ImagePullBackOff   0          3m1s

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get deployments
NAME                       READY   UP-TO-DATE   AVAILABLE   AGE
registration-app-deployment   0/2     2            0           3m15s
```




## Step: 06 Verify get pods, get deployments, ...

```
C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get deployments
NAME                            READY   UP-TO-DATE   AVAILABLE   AGE
registration-app-deployment     0/2     2            0           3m15s

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get svc
NAME                       TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
registration-app-service   NodePort   10.110.170.199   <none>        80:31713/TCP   3m13s

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get all
NAME                                                READY   STATUS         RESTARTS   AGE
pod/registration-app-deployment-785b95ffc5-skt9g    0/1     ErrImagePull   0          3m29s
pod/registration-app-deployment-785b95ffc5-smrqq    0/1     ErrImagePull   0          3m29s

NAME                               TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/registration-app-service   NodePort   10.110.170.199   <none>        80:31713/TCP   3m20s

NAME                                           READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/registration-app-deployment    0/2     2            0           3m29s

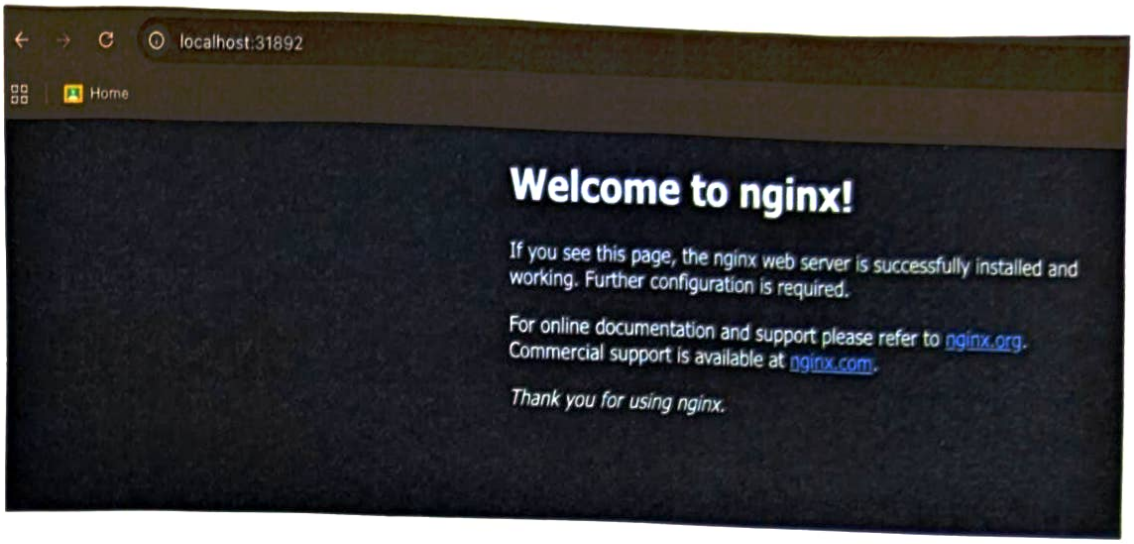
NAME                                                      DESIRED   CURRENT   READY   AGE
replicaset.apps/registration-app-deployment-785b95ffc5    2         2         0       3m29s

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>minikube service registration-app-service
'minikube' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get svc registration-app-service
NAME                       TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
registration-app-service   NodePort   10.110.170.199   <none>        80:31713/TCP   4m26s

C:\Users\varsh\OneDrive\Desktop\simple-k8s-app>kubectl get svc registration-app-service
NAME                       TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
registration-app-service   NodePort   10.110.170.199   <none>        80:31713/TCP   4m45s
```

Output:-


localhost:31892
Home
Welcome to nginx!
If you see this page, the nginx web server is successfully installed and working. Further configuration is required.
For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.
Thank you for using nginx.

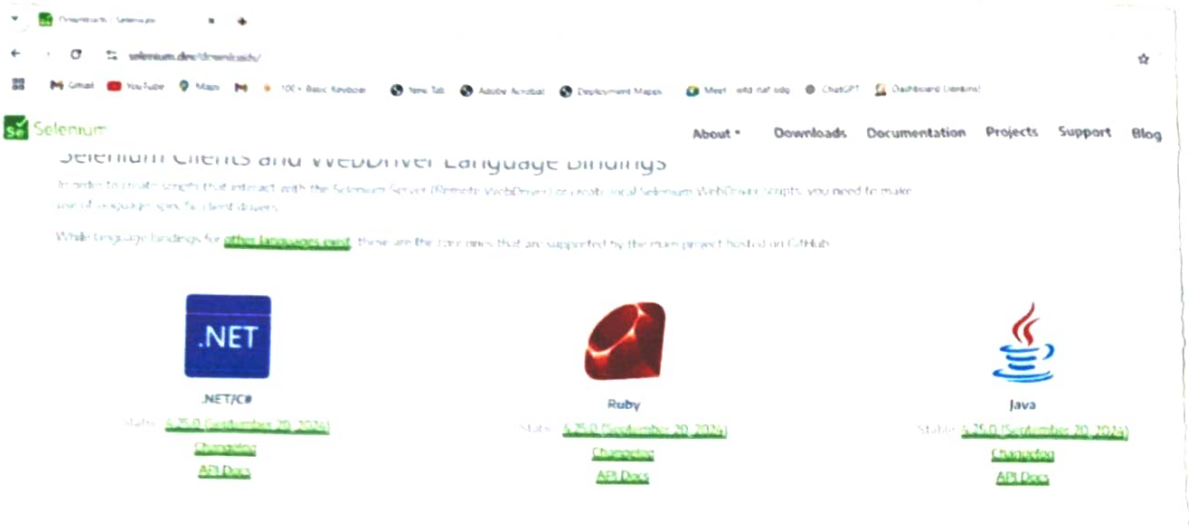
Experiment : Selenium Date:

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# Exp:10 Install and Explore Selenium for Automated Testing.

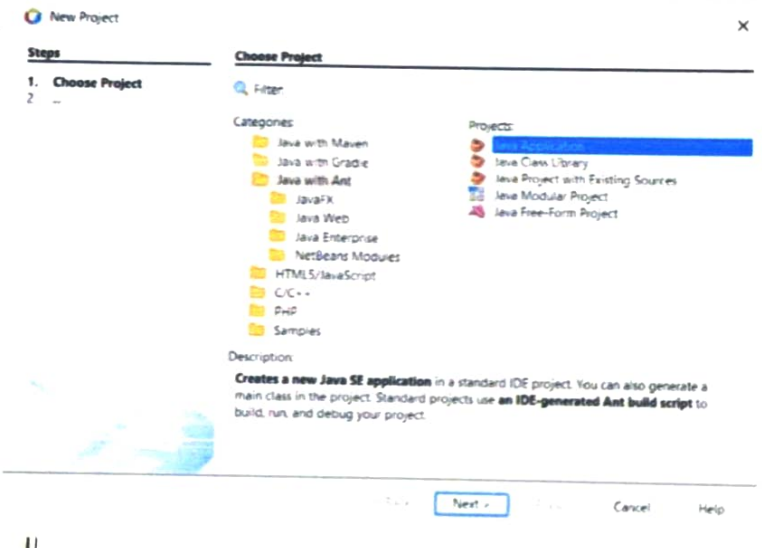
Install netbeans IDE and JDK 17

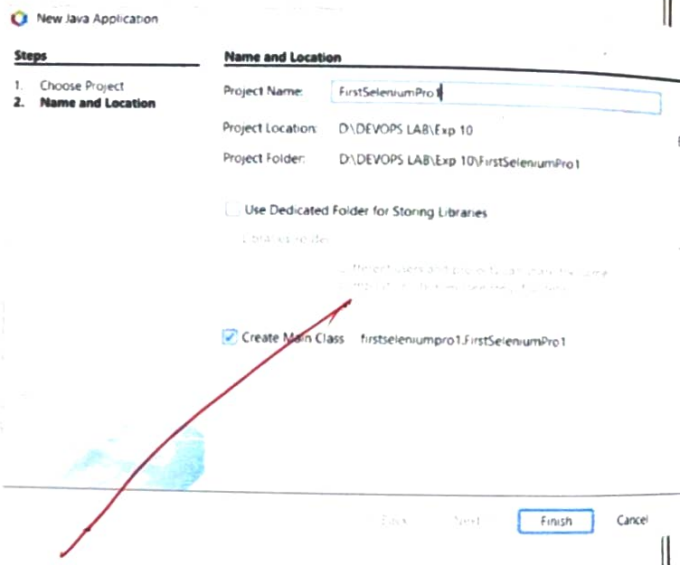
Install Jar files from the Selenium Downloads.




Unzip your cooking directory.

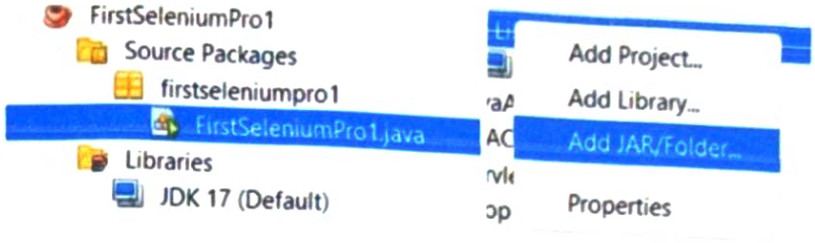
Open Netbeans, Go to files → New Project.



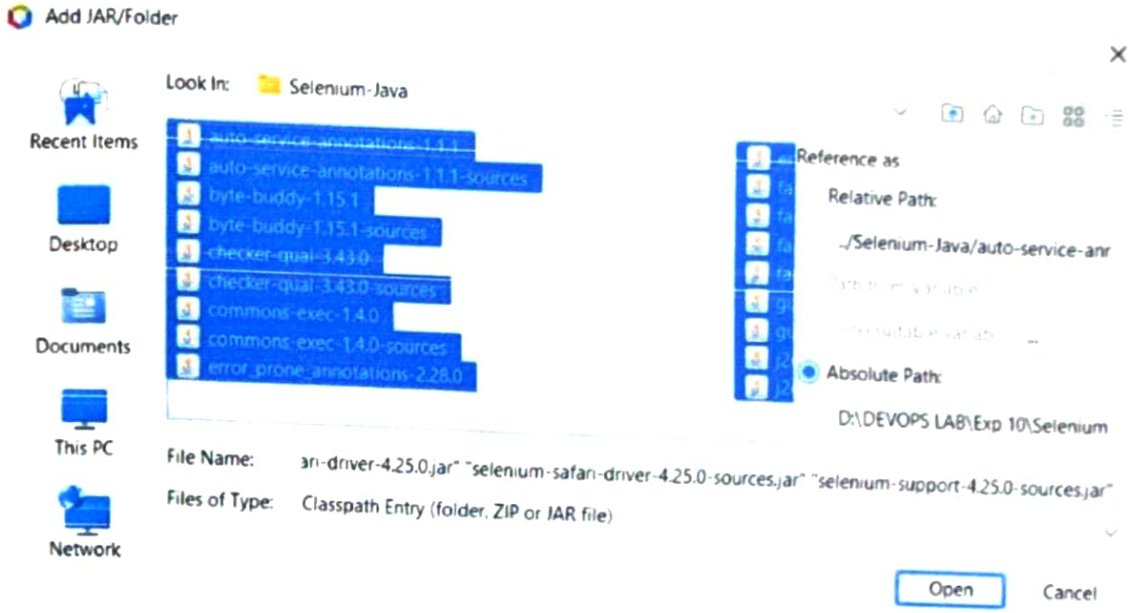




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Right click on Libraries and select ADD JAR Files

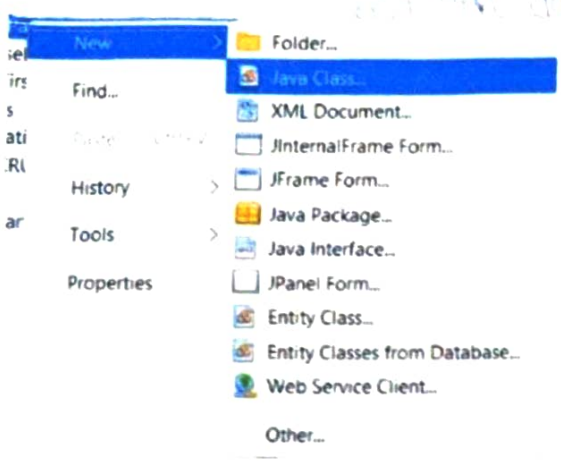


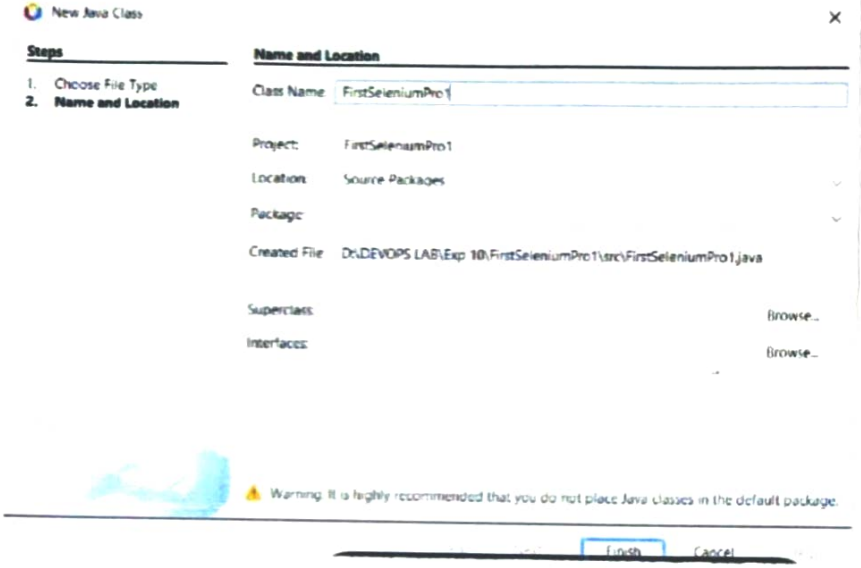

Add all JAR files by clicking ctrl+A




Create a 'javaclass file and give it a name & class.

Click on Finish.


New
Find...
History
Tools
Properties
Folder...
Java Class...
XML Document...
JInternalFrame Form...
JFrame Form...
Java Package...
Java Interface...
JPanel Form...
Entity Class...
Entity Classes from Database...
Web Service Client...
Other...


New Java Class
Steps
1. Choose File Type
2. Name and Location
Name and Location
Class Name: FirstSeleniumPro1
Project: FirstSeleniumPro1
Location: Source Packages
Package:
Created File: D:\DEVOPS LAB\Exp 10\FirstSeleniumPro1\src\FirstSeleniumPro1.java
Superclass:
Interfaces:
Browse...
Browse...
Warning: It is highly recommended that you do not place Java classes in the default package.
Finish
Cancel

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Experiment : _____ Date: _____

Roll No.: 5G1 Page No.: 67

FirstSeleniumProl.java

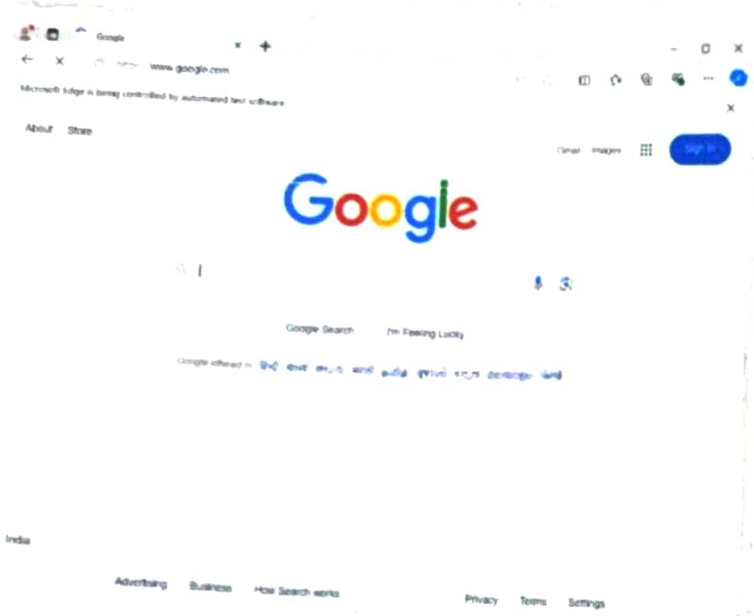
```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.edge.EdgeDriver;
public class FirstSeleniumProl {
  public static void main(String[] args) {
    System.setpropely("webdriver.edge.driver", "c:\\Users\\.exe");
    WebDriver driver = new EdgeDriver();
    try {
      driver.get("https://www.google.com");
      System.out.println("Title:" + driver.getTitle());
      Thread.sleep(60000); }
    catch(Exception e) {
      System.out.println("Error:" + e.getMessage());
    finally {
      driver.quit();
      }
    }
}
```

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Experiment : _____ Date: _____

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Output:-




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Experiment : Event Registration. Date:

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1) Open NetBeans, Go to File → New Project

2) Give Project Name → Event Registration by clicking on Java with Ant → Java Application.

3) After creating project, right click on Libraries and select on Add JAR File.

4) Add all Jar files by clicking Ctrl + A.

5) Create a java class file and give it name & class → Event Registration.

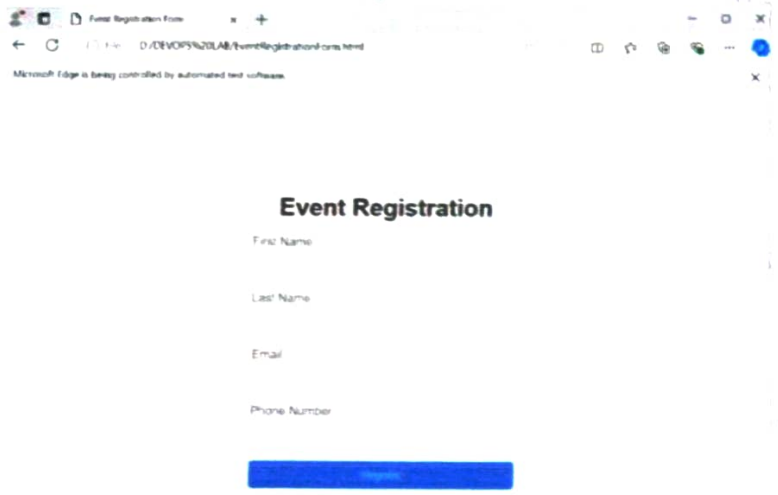
EventRegistration.java

```
import org.openqa.selenium.WebDrivers;
import org.openqa.Selenium.edge.EdgeDriver;
public class EventRegistration {
  public static void main(String[] args) {
    System.setPropery("WebDriver.edge.driver", "c:\\Users\\exe");
    WebDriver driver = new EdgeDriver();
    try {
      System.out.println("Launching");
      driver.get("file:///C:Users/EventRegistration.html");
      System.out.println("Title: "+driver.getTitle());
      Thread.sleep(60000); }
```

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```
catch (Exception e) {
    System.out.println("Error: " + e.getMessage()); }
finally {
    driver.quit(); } } }
```

Output:-



Experiment : _____ Date: _____

Roll No.: 561 Page No.: 71

Exp 11:- Write a simple program in Java Script and perform testing using selenium.

Calculator.java

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.edge.EdgeDriver;
public class Calculator {
 public static void main(String[] args) {
  System.setPropely("webdriver.edge.driver","c:\\Users\\-exe");
  WebDriver driver = new EdgeDriver();
  try{
   System.out.println("Launching");
   driver.get("C:// Users // Calculator.html");
   System.out.println("Title:" +driver.getTitle());
   Thread.sleep(60000); }
catch(Exception e){
  System.out.println("Error:" +e.getMessage()); }
finally {
  driver.quit();
  }
 }
}
```

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Calculator.html

```
<!DOCTYPE html>
<html>
<head>
 <title> Calculator </title>
 <style>
  body {
    font-family : Arial;
    text-align : center;
    margin-top : 50px;
  }
  input {
    width : 200px;
    height : 30px;
    font-size : 18px;
  }
  button {
    width : 50px;
    height : 40px;
  }
 </style>
</head>
```

```
<body>
<input type = "text" id = "display" readonly><br>
<button onclick = "appendValue('1')">1</button>
<button onClick = "appendValue('2')">2</button>
<button onclick = "appendValue('3')">3</button><br>

<button onClick = "appendValue('4')">4</button>
<button onclick = "appendValue('5')">5</button>
<button onClick = "appendValue('6')">6</button><br>

<button onclick = "appendValue('7')">7</button>
<button onclick = "appendValue('8')">8</button>
<button onclick = "appendValue('9')">9</button><br>

<button onclick = "appendValue('0')">0</button>
<button onClick = "appendValue('+')">+</button>
<button onClick = "appendValue('-')">-</button><br>

<button onclick = "appendValue('*')">*</button>
<button onclick = "appendValue('/')">/</button>
<button onclick = "calculate()">=</button><br>

<script>
function appendValue(val){
```

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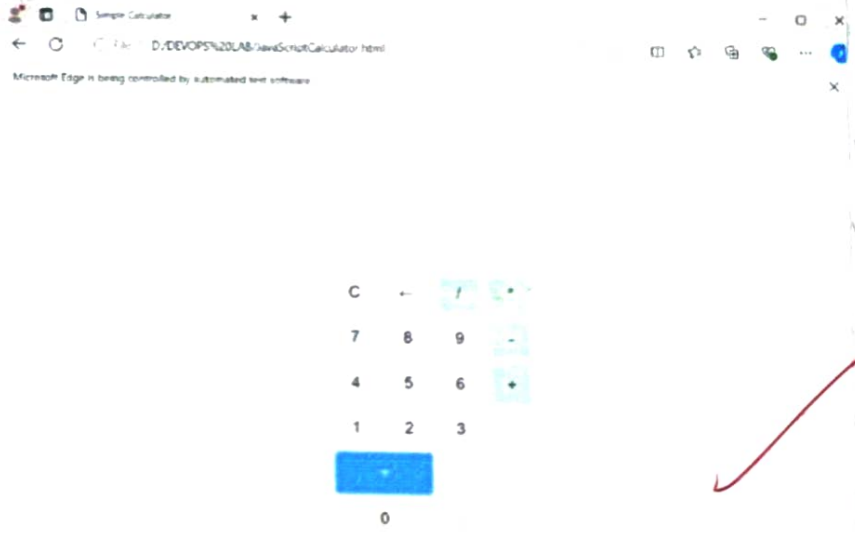
```
    document.getElementById("display").value += val; }

function calculate() {
    let result = eval(document.getElementById("display").value);
    document.getElementById("display").value = result; }

function clearDisplay() {
    document.getElementById("display").value = ""; }

</script>
</body>
</html>
```

Output:



Experiment : 10. Test cases for container application in Selenium. Date:

Roll No.: 5G1 Page No.: 75

Develop test cases for the above containerized application using Selenium

1) Create a folder my-regi-form.

2) Folder structure

↳ deployment

↳ Dockerfile

↳ main

↳ msedgedriver

↳ Sample Script

↳ service

↳ style

↳ success

regi.html

```
<!DOCTYPE html>
<html>
<head>
  <title> Registration Form </title>
  <link rel = "stylesheet" href = "style.css">
  <script defer src = "main.js" > </script>
</head>
<body>
  <div class = "container">
```

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```
<h2> Registration Form </h2>
<form id = "regForm">
<label> Username: </label>
<input type = "text" id = "username">
<span class = "error" id = "usernameError"></span>
<label> Email: </label>
<input type = "email" id = "emailError">
<span class = "error" id = "emailError"></span>
<label> Password: </label>
<input type = "password" id = "password">
<span class = "error" id = "passwordError"></span>
<label> Confirm Password </label>
<input type = "password" id = "Confirm Password">
<span class = "error" id = "Confirm Error"></span>
<button type = "submit" id = "RegisterBtn"> Register </button>

</form>
</div>
</body>
</html>
```

success.html

```
<!DOCTYPE html>
<html>
<head>
  <title> Registration Successful </title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <div class="success-box">
  <h2> Registration Success </h2>
  <a href="reg.html" class="back-btn"> Go Back </a>
</div>
</body>
</html>
```

Main.js

```
document.getElementById("regForm").addEventListener("submit", function(event){
    event.preventDefault();
document.getElementById("usernameError").innerText = "";
document.getElementById("emailError").innerText = "";
document.getElementById("passwordError").innerText = "";
```

```
let valid = true;
let username = document.getElementById("username").value.trim();
let email = document.getElementById("email").value.trim();
let password = document.getElementById("password").value.trim();

if(username === "") {
  document.getElementById("usernameError").innerText = "Username req";
  valid = false; }

if(email === "") {
  document.getElementById("emailError").innerText = "Email is required";
  valid = false; }

else if(password !== confirmPassword) {
  document.getElementById("confirmPasswordError").innerText = "Password not match
  valid = false;

}
 if(valid) {
   window.location.href = "success.html";

   }

});
```

Deployment.yaml

```
apiVersion : apps/v1
kind: Deployment
metadata :
  name : my-regi-form
  labels :
    app : my-regi-form

spec :
  replicas : 2
  selector :
    matchLabels :
      app : my-regi-form
  template :
    metadata :
      labels :
        app : my-regi-form

    spec :
      containers :
        - name : my-regi-form
          image : my-regi-form/latest
          ports :
            - containerPort : 8084
```

Services.yaml.

```
apiVersion: v1
kind: Service
metadata:
  name: my-regi-form-service
spec:
  selector:
    app: my-regi-form
  type: NodePort
  ports:
    -protocol: TCP
     port: 8084
     targetPort: 8084
     nodePort: 30084
```

Dockerfile:

```
FROM nginx:alpine
COPY regi.html /user/share/nginx/html/regi.html
COPY style.css/usr/share/nginx/html/style.css
COPY main.js/usr/share/nginx/html/main.js
COPY success.html/usr/share/nginx/html/success.html
```

Sample Script.py

```
import time
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.edge.service import Service
print("Test Execution started")
driver_path = r"C:\Users\msedgedriver.exe"

options = Options()
options.add_argument('--ignore-ssl-errors=yes')
options.add_argument('-ignore-certificate-errors')
options.add_argument('--disable-logging')

service = Service(driver_path)
driver = WebDriver.Edge(service=service, options=options)

try:
 driver.get("http://localhost:8084/regi.html")
 username = WebDriverWait(driver, 10).until(
   EC.presence_of_element_located((By.ID, "username")) )

username.send_keys("testuser").
```

```
email = driver.find_element(By.ID, "email")
email.send_keys("test@mail.com")
password = driver.find_element(By.ID, "password")
password.send_keys("Test@123")

register = driver.find_element(By.ID, "regisBtn")
register.click()

time.sleep(4)

except Exception as e:
  print("Error occurred:", e)

finally:
  driver.quit()
```

Steps to Run:-

1) pip install selenium

docker pull selenium/standalone-chrome

docker build -t my-regi-app .

docker run -d -p 8084:80 my-regi-form

Output:




**Registration Successful!**

You have registered successfully.

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```
PS C:\Users\BDALAB-33\Desktop\my-regi-form> python SampleScript.py
Test Execution Started
Navigated to registration form...
Form submitted successfully!
Test Execution Finished
```